Module 1 | Introduction to Data Analytics

From Data → Insights → Decisions

Data Analytics Course
Dr. Rehab Mahmoud
rma08@fayoum.edu.eg
IS department, Faulty of computer sciences and AI
Fayoum university

Learning Objectives

Understand what Data Analytics means and why organizations use it.

Differentiate Data Analysis from Data Analytics.

Recognize the four major types of analytics.

Understand the Data Analytics Lifecycle.

Translate a business problem into an analytical question.

Identify common roles, skills, and tools in a data analytics career.

What Is Data?

Data = raw facts, observations, measurements, or records.

Examples:

- Sales transactions: date, product, quantity, price
- Customer data: age, location, segment, purchase history
- Website data: visits, clicks, sessions, conversion
- Operational data: inventory, delivery time, support tickets

Data becomes valuable when we organize, analyze, and interpret it.

From Data to Decision



Example: 10,000 transactions are data. Discovering that one customer segment drives 45% of revenue is an insight. Deciding where to focus marketing is a decision.

What Is Data Analytics?

A systematic process of examining data to discover patterns, trends, relationships, and actionable insights.

Its purpose is not only to describe numbers—but to support better questions, decisions, and actions.

Typical flow:

- Collect / access data
- Clean & prepare
- Analyze
- Visualize & communicate
- Recommend / support decisions

Data Analysis vs. Data Analytics

	Data Analysis	Data Analytics
Focus	Examining data to answer questions and extract insights.	Broader discipline covering data preparation, analysis, visualization, statistics, prediction, and decision support.
Typical question	What happened? Why did it happen?	What happened? Why? What may happen? What action could be considered?
Scope	Often refers to the analytical activity itself.	Often refers to the wider process, capability, or field.

The Four Types of Analytics

1. Descriptive

What happened?
Sales decreased by 12% last month.

2. Diagnostic

Why did it happen?
The decrease came mainly from Region A.

3. Predictive

What may happen?
Sales may continue to decline next month.

4. Prescriptive

What could we do?
Increase promotion in the affected segment.

1. Descriptive Analytics

Summarizes historical or current data.

Answers: “What happened?”

Common outputs:

- KPIs and metrics
- Reports
- Dashboards
- Trends and summaries

Examples: total revenue, monthly sales, average order value, customer count.

2. Diagnostic Analytics

Investigates causes and relationships behind an observed result.

Answers: “Why did it happen?”

Common techniques:

- Drill-down analysis
- Segmentation
- Comparison
- Correlation analysis

Example: Sales fell → compare regions → identify a sharp decline in one product category.

3. Predictive Analytics

Uses historical data, patterns, and statistical / machine-learning methods to estimate future outcomes.

Answers: “What may happen?”

Examples:

- Forecasting future sales
- Predicting customer churn
- Estimating demand
- Risk prediction

Prediction is probabilistic—not a guarantee of the future.

4. Prescriptive Analytics

Uses analytical results to evaluate possible actions and support decisions.

Answers: “What could we do?”

Examples:

- Which products should be promoted?
- How should inventory be allocated?
- Which customers should receive an offer?

May use optimization, simulation, rules, or scenario analysis.

Data Analytics Lifecycle

Business Understanding

Data Collection

Data Preparation

Exploration & Analysis

Visualization

Insights & Communication

Action / Evaluation

Business Question → Analytical Question

Business question: “Why are our sales declining?”

Break it into analytical questions:

- Which products declined?
- Which regions / channels declined?
- When did the decline start?
- Which customer segments were affected?
- Is the change statistically / commercially meaningful?

Good analytics starts with a clear question—not with a tool.

Mini Case Study: E-Commerce

Situation: An online store notices that monthly revenue dropped 10%.

Descriptive → Which months, products, and channels changed?

Diagnostic → What factors are associated with the decline?

Predictive → What could next month look like if the trend continues?

Prescriptive → Which actions could improve revenue?

Discussion: What data would you need to answer these questions?

Data Analytics Career Path

Data Analyst
Reports, SQL, Excel, dashboards, insights

BI Analyst
Business intelligence, Power BI, KPIs, reporting

Product Analyst
User behavior, product metrics, experiments

Marketing Analyst
Campaigns, customers, conversion, ROI

Business Analyst
Business requirements, processes, decision support

Data Scientist
Advanced statistics, ML, predictive modeling

Core Skills of a Data Analyst

Technical Skills: Excel, SQL, Power BI, Python, statistics.

Analytical Skills: problem solving, critical thinking, pattern recognition.

Business Skills: understanding KPIs, processes, customers, and objectives.

Communication Skills: explaining insights clearly to non-technical audiences.

Data Literacy: understanding data quality, limitations, bias, and context.

Curiosity: asking better questions before jumping to conclusions.

Tools in the Data Analytics Ecosystem

Excel
Quick analysis & business spreadsheets

SQL
Querying databases

Power BI
BI & interactive dashboards

Python
Automation, EDA & advanced analysis

Statistics
Uncertainty & evidence

AI Tools
Assistance, productivity & exploration

Class Activity

Choose one business scenario: Retail / Hospital / University / Bank / E-commerce.

Write one business problem.

Convert it into 3 analytical questions.

Classify each question as Descriptive, Diagnostic, Predictive, or Prescriptive.

Identify 3–5 data fields you would need.

Module 1 | Key Takeaways

Data is raw; analytics turns data into useful evidence and insights.

Data Analysis is commonly used for examining data; Data Analytics is a broader discipline.

Four types: Descriptive → Diagnostic → Predictive → Prescriptive.

A strong analytics process begins with a business question.

Tools are important, but analytical thinking and communication are equally important.

Next Module

Excel for Data Analytics

From raw spreadsheets to clean, analysis-ready data